Operational Scenarios: Vegetative Growth, Flowering, and Harvesting

Hydroponic IoT System for Leafy Greens

SCENARIO 1: VEGETATIVE GROWTH STAGE

System Preparation:

Nutrient reservoir filled to 80% capacity (15L for balcony system)

Primary nutrients mixed: N-P-K ratio 3-1-2 for leafy greens

pH adjusted to 5.8 ± 0.2 using automated dosing system

EC/TDS set to 1.2-1.8 mS/cm for seedling stage

Water temperature stabilized at 20°C ± 1°C

Seedling Integration:

12-15 mature seedlings (3-4" tall) placed in net pots

Root systems carefully inserted into growing channels

Spacing: 15-20cm between plants for airflow

Rockwool/coco coir cubes pre-soaked in pH-balanced water

IoT System Configuration:

Plant profile loaded: "Leafy Greens - Vegetative Stage"

Target parameters set:

Temperature: 20-24°C (Day), 18-20°C (Night)

Humidity: 60-70% RH

pH: 5.5-6.2

EC: 1.2-1.8 mS/cm

Light: 16 hours ON, 8 hours OFF

Light Intensity: 250-400 µmol/m²/s

CO2: Ambient (400-600 ppm)

Nutrient Pump: 15 min ON / 45 min OFF cycle

Daily Operations (Days 1-14)

Morning Cycle (06:00 - 12:00):

06:00: Lights ON (gradual ramp to 70% intensity over 30 minutes)

06:05: Water pump activates - 15-minute nutrient circulation

06:20: System performs morning sensor readings

- pH: 5.8 (within range)

- EC: 1.5 mS/cm (optimal)

- Water temp: 20.5°C

- Air temp: 22°C

- Humidity: 65%

- Reservoir level: 75%

06:30: Axial fan activates at 50% speed for air circulation

08:00: First nutrient circulation of the day

10:00: System check - all parameters logged to cloud

10:05: If humidity >70%, fan speed increases to 75%

Afternoon Cycle (12:00 - 18:00):

12:00: Light intensity increases to 100% (400 µmol/m²/s)

12:05: Water pump circulation (15 minutes)

14:00: Temperature monitoring peak

- If air temp > 24°C: Fan speed to 100%

- If water temp > 22°C: LED intensity reduced to 80%

15:00: Nutrient concentration check

- EC drops below 1.2 mS/cm: Alert sent: "Add nutrients A+B"

- EC above 1.8 mS/cm: Add RO water, alert sent

16:00: pH adjustment cycle

- If pH > 6.2: Micro-dose pH Down (0.1ml)

- If pH < 5.5: Micro-dose pH Up (0.1ml)

Evening/Night Cycle (18:00 - 06:00):

18:00: Lights begin 30-minute ramp-down to 30% intensity

18:30: Water pump final circulation (10 minutes)

19:00: Lights OFF

Night parameters activate:

- Temperature target: 18-20°C

- Humidity: 65-75% (passive)

- All pumps idle except for emergency circulation

22:00: Midnight sensor reading (once per night)

Data transmitted to cloud for daily analytics

03:00: Optional: Brief 5-minute circulation if root zone temp > 20°C

Weekly Maintenance (Automated + User Tasks)

System Day 7: First Major Check

***IoT Automated Tasks**:*

1. Nutrient solution refresh recommendation

- "Reservoir at 60%, recommend partial change in 3 days"

2. Root health assessment via camera (if equipped)

- Color analysis: White/cream roots = healthy

- Alert if browning detected

3. Growth rate calculation:

- Daily increase: 1.2cm (optimal)

- Leaf count: Increased from 4 to 8 true leaves

User Tasks (15 minutes):

1. Visual inspection of plants

2. Clean any algae from net pots

3. Top up reservoir with pH-balanced water

4. Record observations in app

System Day 14: Mid-Veg Transition

System Adjustments:

1. Increase EC to 1.8-2.2 mS/cm

2. Adjust nutrient ratio to 4-2-3 (higher nitrogen)

3. Increase light intensity to 450 µmol/m²/s

4. Extend photoperiod to 17 hours if growth < target

IoT Alerts:

- "Vegetative growth phase halfway complete"

- "Plants showing 12-15 true leaves"

- "Root mass increased by 300%"

- "Reservoir consumption: 8L/week"

IoT Data Flow During Vegetative Stage

Sensors → ESP32 → Cloud → User Dashboard

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Real-time Parameters:

1. VPD (Vapor Pressure Deficit): 0.8-1.1 kPa

2. DLI (Daily Light Integral): 14-17 mol/m²/day

3. Nutrient uptake rate: 0.8-1.2 mS/cm decrease per week

4. Water consumption: 1.1-1.5L/day

5. Temperature differential (air/water): 2-4°C

Automated Decisions:

1. Nutrient dosing based on EC trendline

2. pH correction based on drift patterns

3. Light adjustment based on DLI accumulation

4. Fan control based on VPD calculation

SCENARIO 2: FLOWERING/BOLTING PREVENTION & MATURATION

Transition to Maturation (Days 15-21)

System Override: "Prevent Bolting Protocol"

1. Temperature Control:

- Strict maintenance at 18-22°C

- Night temp never below 16°C to avoid cold stress

- Day temp never above 24°C to avoid heat stress

2. Light Management:

- Maintain 16-hour photoperiod (DO NOT increase)

- Light intensity: 400-500 µmol/m²/s

- No red spectrum increase (prevents flowering signals)

3. Nutrient Adjustment:

- Reduce nitrogen slightly (3-1-3 ratio)

- Maintain EC: 1.8-2.0 mS/cm

- Add calcium supplement (5ml/10L) for leaf strength

4. Stress Prevention:

- Consistent water cycles (no dry periods)

- Stable pH (5.8-6.0)

- Minimal parameter fluctuations

Daily Monitoring for Bolting Signs

IoT Detection Algorithms:

1. Plant Height Monitoring:

- Expected: 0.8-1.2cm/day growth

- Alert: >2cm/day elongation = possible bolting

2. Internode Spacing Analysis:

- Normal: 2-3cm between leaf nodes

- Warning: >4cm spacing = stretching/bolting

3. Leaf Morphology:

- Normal: Broad, flat leaves

- Bolting: Leaves become pointed, smaller

- IoT camera analysis for shape changes

4. Apical Dominance Check:

- Center growth point monitoring

- Alert if central stem elongates rapidly

Bolting Intervention Protocols

Level 1 Alert (Early Signs):

- "Possible bolting detected - adjusting environment"

Actions:

1. Reduce temperature by 2°C (to 18-20°C)

2. Increase air circulation (fan to 80%)

3. Slight reduction in light intensity (10%)

4. Ensure consistent watering

Level 2 Alert (Confirmed Bolting):

- "Bolting detected - immediate harvest recommended"

Actions:

1. System prepares for harvest

2. Flush cycle initiated (plain water for 24-48 hours)

3. Nutrient dosing stopped

4. User alerted for immediate action

Pre-Harvest Flush (Final 2-3 Days)

1. Reservoir drained to 20%

2. Refilled with reverse osmosis water + flushing agent

3. pH adjusted to 6.0

4. EC reduced to 0.4-0.6 mS/cm

IoT Monitoring During Flush:

1. pH stability check (every 4 hours)

2. Water temperature maintained at 20°C

3. Regular circulation (10min ON/30min OFF)

4. Final nutrient uptake measurement

Flush Complete Criteria:

1. Runoff EC < 1.0 mS/cm

2. 48 hours elapsed

3. Leaf color bright green (no dark green/blue tinge)

HARVESTING & SYSTEM RESET

IoT Harvest Indicators (All must be met):

1. Time-Based: 28-35 days from transplant

2. Size-Based:

- Plant height: 20-25cm (for lettuce)

- Leaf count: 12-16 true leaves minimum

- Canopy coverage: 80% of grow area

3. Physiological:

- Growth rate slowed to <0.5cm/day

- No new leaf emergence in 48 hours

- Lower leaves beginning to yellow slightly

4. Environmental:

- Flush completed (EC < 1.0 mS/cm)

- 24-hour dark period completed (optional)

Harvest Alert to User:

"Harvest window open: Next 3 days optimal

Current status:

- Days since transplant: 32

- Average plant height: 23cm

- Canopy coverage: 85%

- Flush completed: Yes

- Recommended harvest time: Morning"

Harvest Day Operations

Morning of Harvest (User + System Coordination):

Pre-Harvest (06:00 - 08:00):

1. System maintains dark period (lights OFF)

2. Water temperature lowered to 18°C

3. Final circulation (5 minutes)

4. Reservoir drained to collection container

Harvest Window (08:00 - 10:00):

1. User removes plants from system

2. Roots separated from stems

3. Harvest weight recorded in app

4. Quality assessment:

- Leaf color (1-5 scale)

- Leaf texture (crispness test)

- Root health (white/brown ratio)

Post-Harvest System Actions:

1. Water pump runs cleaning cycle (water + hydrogen peroxide)

2. All sensors take baseline readings

3. System logs harvest data:

- Total yield: 450-600g per square foot

- Water consumed: 25-30L total

- Electricity used: 45-50 kWh

- Days to harvest: 32

System Reset for Next Cycle

Phase 1: Deep Clean (2-4 hours)

1. Drain all fluid from system

2. Circulate cleaning solution (3% H_2O_2) for 1 hour

3. Rinse with pH-balanced water (3 cycles)

4. UV sterilization (if equipped) for 30 minutes

Phase 2: System Check (IoT Automated)

1. Sensor calibration:

- pH sensor: 4.0 and 7.0 buffer check

- EC sensor: 1.41 mS/cm calibration solution

- Temperature: Cross-check with calibrated thermometer

2. Component testing:

- Pump flow rate verification

- LED spectrum/output test

- Fan speed calibration

- Relay switching test

3. Safety checks:

- Ground continuity test

- Leak detection test

- Over-temperature protection test

Phase 3: Preparation for Next Batch

1. Refill reservoir with fresh nutrients

2. Load next plant profile

3. Pre-soak growing media

4. Set countdown to seedling arrival

5. Send notification: "System ready for next batch in 24 hours"

Post-Harvest Analytics & Optimization

IoT-Generated Harvest Report:

1. Yield Analysis:

- Total fresh weight: 2.1kg

- Yield per kWh: 42g/kWh

- Yield per liter: 70g/L

- Growth rate: 65g/day

2. Efficiency Metrics:

- Water use efficiency: 14L/kg

- Nutrient efficiency: Cost per gram

- Space-time yield: 1.8kg/m²/month

3. Quality Assessment:

- Leaf nitrate levels (estimated from EC history)

- Vitamin content prediction (from light spectrum data)

- Shelf life estimate: 7-10 days based on growth conditions

4. Recommendations for Next Cycle:

- "Increase light intensity by 10% during weeks 2-3"

- "Reduce night temperature by 1°C"

- "Begin flush 12 hours earlier"

- "Consider 10% higher plant density"

Operational Parameters Matrix:

Phase	Duration	Temp (°C)	Humidity	pH	EC (mS/cm)	Light (hrs)	Notes
Early Veg	7-10d	20-22	65-75%	5.8-6.0	1.2-1.6	16	Seedling establishment
Mid Veg	10-14d	21-23	60-70%	5.8-6.2	1.6-2.0	16-17	Rapid growth phase
Late Veg	7-10d	18-22	55-65%	5.8-6.0	1.8-2.2	16	Maturation, bolt prevention
Flush	2-3d	18-20	50-60%	6.0-6.2	0.4-0.8	0-12	Nutrient clearing

IoT Control Logic Summary

IF temperature > setpoint +2°C THEN fan = 100%

IF temperature < setpoint -2°C THEN heater = ON

IF humidity > setpoint +10% THEN fan = +25%

IF EC < threshold -0.3 THEN alert "Add nutrients"

IF pH > 6.2 THEN dose pH Down (0.05ml/L)

IF water level < 30% THEN alert "Refill reservoir"

IF power loss > 5min THEN activate emergency protocol

Success Metrics for Each Scenario:

1. **Vegetative Growth:** 85% survival rate, >1cm/day growth, no nutrient deficiencies
2. **Bolting Prevention:** <5% bolting incidence, uniform canopy development
3. **Harvest:** >400g/sq ft yield, <35 days to harvest, >7-day post-harvest shelf life
4. **System Reset:** <6 hours turnaround, 100% component functionality verified